already applied to atomic nuclei, explain the fact that this surface tension causes these drops to assume a spherical shape. For the surface energy of a drop is proportional to the area of the surface, and it consequently tends to make the surface area as small as possible. We must assume the presence of such surface tension in atomic nuclei, too. As a result of the lesser cohesion of the particles situated on the surface, this surface tension must produce a decrease of the total binding energy, and consequently, of the average energy per particle as well. Just as in a liquid drop, the surface tension is the cause of the spherical shape of atomic nuclei, too.

There is, however, one essential difference between nuclear matter and a liquid. The liquid consists of molecules which are electrically neutral, whereas the nuclear matter contains not only neutrons but electrically charged protons, too. Therefore, our analogy must be one between nuclei and liquid drops containing electrically charged molecules with forces of repulsion operating between them. There is also an electric force of repulsion present in the nuclei of atoms.

III. THE THREE TYPES OF NUCLEAR ENERGY

We may therefore regard the energy content of a nucleus as the sum total of three components. The major part of this energy comes from the nuclear forces which make the cohesion of the nucleus possible. This energy is modified by the surface tension. Finally, a part of this total energy originates from the force of electric repulsion. Let us consider these three components separately and correlate them with the number of protons and neutrons on the basis of a study by v. Weizsäcker.

We shall begin with the nuclear forces. These are the forces which bind the protons and neutrons together, and as we have seen, they are related to the fact that nuclei are capable of emitting both electrons and positrons. This phenomenon is evidently quite symmetrical as between protons and neutrons. A neutron can change into a proton, in which process an electron is emitted, and conversely, a proton can change into a neutron, in which case a positron is the by-product of the process. This fact warrants the conclusion that with respect to the nuclear forces, or to the *nuclear field,* there is no difference between protons and neutrons. Therefore, it must be possible to represent that part of the binding energy which is the product of the nuclear forces, by some *symmetric* function of the number of neutrons and protons. Now, if we first write this function in a general form, and then, near the point where the number of neutrons is equal to the number of protons, we stop with the second term, we obtain a simple equation for the binding energy per particle (in so far as this energy originates from the nuclear field), as follows: